

Recommendation letter  
for  
Kristiyan Ivanov Garchev  
by  
Assoc. Professor Parvan Parvanov, PhD  
Interim Rector of Sofia University "St. Kliment Ohridski"  
Department of Mathematical Analysis  
Faculty of Mathematics and Informatics

To the Admissions Committee,

I am pleased to write this letter in support of Kristiyan Garchev for admission to your graduate program. I know Kristiyan through a substantial body of advanced mathematical coursework in analysis and related fields. His performance across these courses, reinforced by his international distinction in mathematics competitions, places him firmly at the very top of his cohort. He is not only a top-ranking student but one of the strongest young mathematicians our faculty has trained in recent years. He combines exceptional technical ability with genuine mathematical maturity, and this makes him especially well suited for further rigorous study.

My first impression of Kristiyan was formed through foundational analysis courses such as Calculus I, Calculus II, and Math Analysis (Calculus III). Even at this early point in his academic development, he stood out for the seriousness and depth of his engagement with the material. He was not content simply to master standard techniques, but consistently sought the underlying ideas and showed a strong instinct for rigor, structure, and generalization.

This became even clearer in advanced courses such as Complex Analysis, Ordinary Differential Equations, and Partial Differential Equations, where Kristiyan showed a strong ability to balance technical precision with genuine theoretical understanding. He worked carefully with analytical arguments while maintaining a clear view of the broader mathematical structure. The same strength was evident in Introduction to Functional Analysis and Functional Analysis. Colleagues in our department were particularly impressed by how naturally he adapted to the abstract demands of these subjects and by the mathematical depth with which he approached difficult concepts.

His preparation extends further into advanced and structurally demanding subjects such as Homological Algebra, Algebraic Topology, and Convexity and Differentiability in Banach Spaces. These are master's-level courses rather than standard undergraduate offerings, and it is highly unusual to see them pursued so successfully at his stage of study. What is especially impressive is that Kristiyan does not treat such courses as isolated accomplishments, but engages with them in a serious and substantive way. He has a strong instinct for how different areas of mathematics connect, and that is one of the clearest indicators of future academic potential.

Kristiyan's mathematical competition record strongly reinforces this impression. He has distinguished himself in numerous national and international competitions, earning top placements and gold medals. Particularly notable is his 1st place finish at the 2025 Student Team Mathematics Competition organized by the Bulgarian Academy of Science. This achievement is especially significant because some of the problems were open-ended, requiring not only technical skill and

speed, but also originality and genuine mathematical creativity. The same sharpness and inventiveness are clearly visible in his university work.

I would also like to emphasize Kristiyan's work as a teaching assistant in my own Calculus course. This gave me the opportunity to observe him not only as an excellent student, but also as a young mathematician taking on real academic responsibility. Teaching foundational analysis is a serious task: it requires clarity, patience, and a deep enough understanding of the subject to present ideas from multiple perspectives and to guide students from intuition toward rigor. Kristiyan handled this role with maturity, reliability, and genuine competence. He explained material thoughtfully, worked well with students, and showed the kind of command that inspires confidence in both his mathematical preparation and his potential as a future academic.

Another important aspect of his profile is the breadth of his intellectual development. Although my own direct knowledge of him is rooted in mathematical analysis, I have observed that he has also built meaningful connections to a plethora of fields in Mathematics and Computer Science. This is a strength, not a distraction. Students with strong preparation in analysis, differential equations, and functional-analytic thinking often adapt especially well to modern quantitative and machine learning-oriented fields, because they are able to understand models, optimization procedures, and high-dimensional methods structurally rather than treat them as black boxes. This is especially important in areas where success depends not only on implementation, but on a deep grasp of approximation, stability, generalization, and representation. Kristiyan clearly belongs in this category.

On a personal level, he is hardworking, ambitious, and intellectually curious. He is not drawn to easy material or superficial accomplishment. He seeks out demanding subjects, persists through difficulty, and approaches his responsibilities seriously. He has the kind of discipline and curiosity that one hopes to see in a future scholar.

On a more practical note, Kristiyan's English proficiency is excellent. This is evident from his participation in international mathematics competitions conducted in English, his experience in English-speaking professional environments, and his research work, including paper drafting and scholarly collaboration in English.

I therefore recommend Kristiyan Garchev with the highest confidence. He has the analytical depth, the discipline, the competition-tested problem-solving ability, and the teaching experience to succeed in a demanding graduate environment. I am certain that he will continue to develop into an exceptional scholar.

Sincerely,  
Assoc. Prof. Parvanov  
Sofia University "St. Kliment Ohridski"